

RICHARD KNOWLES

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SECURITY ENGINEER · AI & CLOUD INFRASTRUCTURE · DEVSECOPS

Infrastructure security engineer and independent SaaS platform developer who builds security tooling from scratch — not just configures it. Hands-on Go/Python engineering, active vulnerability research (Bugcrowd), and compliance-grade operations across production cloud environments. U.S. Marine Corps veteran, now focused on securing the layer where AI agents touch production systems.

TECHNICAL SKILLS

Cloud & IaC: AWS, Azure, Oracle OCI, Terraform, Ansible, Docker, Proxmox, Cloudflare, PKI/TLS, ZFS

Security Operations: Wazuh SIEM, SentinelONE EDR, CrowdSec, Cowrie honeypots, Burp Suite, Nmap, STRIDE threat modeling

Languages & AI Security: Go, Python (asyncio/FastAPI), Bash, TypeScript, Rust · MCP server development, agentic workflow security, prompt-injection defense

SELECTED SECURITY PROJECTS — GITHUB.COM/RICHKNOWLES

- **pct-svcmap (Go):** Concurrent, zero-dependency service discovery and threat-surface mapping for Proxmox clusters; three-level detection fallback (ss → lsof → /proc/net/tcp) with automated flagging of world-accessible risky services (MySQL, Redis, Telnet, FTP).
- **Peacock (Python/FastMCP):** Linux MCP server giving AI agents scoped filesystem access and command execution — home-directory restriction, command allow-listing, and execution timeouts to prevent sandbox escape.
- **POTA4 (Go/Django/Vue):** Hosted multi-tenant RMM SaaS monitoring ~60 endpoints across 6 client sites in Illinois; full security lifecycle — patch management, threat alerting, incident response.
- **AJ Sender (Python/Oracle OCI):** End-to-end production SaaS campaign platform with API authentication and multi-tenant data isolation.

PROFESSIONAL EXPERIENCE

Bugcrowd | Independent Security Researcher — Bug Bounty | *January 2025 – Present*

- Active vulnerability research across public bug bounty programs — web application testing with Burp Suite, reconnaissance automation, and structured vulnerability reporting.

iTWerks — Elgin, IL | Lead Infrastructure Architect & SaaS Platform Engineer | *2013 – May 2026*

- Designed, deployed, and secured compliance-driven environments for ~50 mid-market clients across healthcare, financial, and legal sectors; 99.9% uptime over 13 years with zero client data loss or compromise — every intrusion detected, contained, and remediated.
- Operated multi-tenant Wazuh SIEM and SentinelONE EDR across distributed client workloads; led SOC 2 and GxP audit readiness with rigorous control documentation and evidence collection.
- Led incident response on a live 34-hour Mirai-variant SSH brute-force attack — isolated via Cowrie honeypots and ipset blocklists, performed full network forensics, and submitted structured threat intelligence to CISA and IC3.
- Contained a live ransomware intrusion at an accounting client — isolated the infection, restored from air-gapped backups with ~15 minutes of data loss, and returned the firm to full operation the same day. No ransom paid.
- Deployed Terraform-managed infrastructure and multi-WAN failover firewalls (Cisco, SonicWall, Barracuda); applied STRIDE threat modeling to client architecture reviews and onboarding workflows.

Structured Communications & Anstar Corp — Illinois | Enterprise Project Manager | *2007 – 2013*

- Delivered a \$30M Central DuPage Hospital infrastructure installation under budget and ahead of schedule with full clinical system validation.
- Executed zero-downtime decommissioning and migration of 560+ legacy servers for an NYSE-listed firm via controlled validation pipelines and phased cutovers.

Motorola Solutions — Schaumburg, IL | Senior Systems Engineer IV & Global Standards Chair | *1997 – 2006*

- Founded the Certified Factory Image (CFI) program — a global supply-chain security initiative validating OS images at the factory with Dell, HP, and Lenovo across 100,000+ endpoints in 7 business units.
- Chaired the Global Hardware Standards Committee; authored enterprise disaster recovery and operational continuity protocols for the WSDD division.

EDUCATION & CREDENTIALS

PortSwigger Web Security Academy — BSCP Certification Track (in progress)

College of DuPage — Computer Science & Business · U.S. Marine Corps — Avionics Systems Technician (Veteran)